Standard ECMA-55

Minimal BASIC

January 1978

Contents

Brief History

The first version of the language BASIC, acronym for Beginner's All-purpose Symbolic Instruction Code, was produced in 1964 at the Dartmouth College in the USA. This version of the language was oriented towards interactive use. Subsequently, a number of implementations of the language were prepared, that differed in part from the original one.

In 1974, the ECMA General assembly recognised the need for a standardised version of the language, and in September 1974 the first meeting of the ECMA Committee TC 21, BASIC, took place. In January 1974, a corresponding committee, X3J2, had been founded in the USA.

Through a strict co-operation it was possible to maintain full compatibility between the ANSI and the ECMA draft standards. The ANSI one was distributed for public comments in January 1976, and a number of comments were presented by ECMA.

A final version of the ECMA Standard was prepared at the meeting of June 1977 and adopted by the General Assembly of ECMA on Dec. 14, 1977 as Standrd ECMA-55.

Scope

This Standard ECMA-55 is designed to promote the interchangeability of BASIC programs among a variety of automatic data processing systems. Subsequent Standrds for the same prpose will describe extensions and enhancements to this Standard. Programs conforming to this Standard, as opposed to extensions or enhancements of this Standard, will be said to be written in “Minimal BASIC”.

This standard establishes:

- the syntax of a program written in Minimal BASIC.
- The formats of data and the precision and range of numeric representations which are acceptable as input to a automatic data processing system being controlled by a program written in Minimal BASIC.
- The formats of data and the precision and range of numeric representations which are acceptable as output to a automatic data processing system being controlled by a program written in Minimal BASIC.
- The semantic rules for interpreting the meaning of a program written in Minimal BASIC.
- The errors and exceptional circumstances which shall be detected and also the manner in which such errors and exceptional circumstances shall be handled.

Although the BASIC language was originally designed primarily for interactive use, the Standard describes a language that is not so restricted.

The organisation of the Standard is outlines in Appendix 1. Thet method of syntax specification is explained in Appendix 2.

References

ECMA-6 : 7-Bit Input/Output Coded Character Set, 4th Edition

ECMA-53 : Representation of Source Programs

Definitions

For the purposes of this Standard, the following terms have the meanings indicated.

BASIC

A term applied as a name to members of a special class of languages which possess similar syntaxes and semantic meanings; acronym for Beginner's All-purpose Symbolic Instruction Code.

Batch-mode

The processing of programs in an environment where no provision is made for user interaction.

End-of-line

The characters or indicator which identifies the termination of a line. Lines of three kinds may be identified in Minimal BASIC: program lines, print lines and input reply lines. End-of-line may vary between the three cases and may also vary depending upon context. Thus, for example, an end of input line may vary on a given system depending on the terminal being used in interactive or batch mode.

Typical examples of end-of-line are carriage-return, carriage-return line-feed, and end of record (such as end of card).

Error

A flaw in the syntax of a program which causes the program to be incorrect.

Exception

A circumstance arising in the course of executing a program which results from faulty data or computations or from exceeding some resource constraint. Where indicated certain exceptions (non-fatal exceptions) may be handled by the specified procedures; if no procedure is given (fatal exceptions) or if restrictions imposed by the hardware or operating environment make it impossible to follow the given procedure, then the exception shall be handled by terminating the program.

Identifier

A character string used to name a variable or a function.

Interactive mode

The processing of programs in an environment which permits the user to respond directly to the actions of individual programs and to control the commencement and termination of these programs.

Keyword

A character string, usually with the spelling of a commonly used or mnemonic word, which provides a distinctive identification of a statement or a component of a statement of a programming language.

The keywords in Minimal BASIC are: BASE, DATA, DEF, DIM, END, FOR, GO, GOSUB, GOTO, IF, INPUT, LET, NEXT, ON, OPTION, PRINT, RANDOMIZE, READ, REM, RESTORE, RETURN, STEP, STOP, SUB, THEN, and TO.

Line

A single transmission of characters which terminates with an end-of-line.

Nesting

A set of statements is nested within another set of statements when:

- the nested set is physically contiguous, and
- the nesting set (divided by the nested set) is non-null.

Print zone

A contiguous set of character positions in a printed output line which may contain an evaluated print statement element.

Rounding

The process by which the presentations of a value with lower precision is generated froma representation of higher precision taking into account the value of that portion of the original number which is to be omitted.

Significant digits

The contiguous sequence of digits . . .

> NOTE: The Standard requires that the ability of a conforming implementation to accept numeric representations be measured in terms of significant digits rather than the actual number of digits (that is including leading or trailing zeroes) in the representation.

Truncation

The process by which the representation of a value with lower precision is generated from a representation of higher precision by merely deleting the unwanted low order digits of the original representation.

Characters and Strings

General Description

The character set for BASIC is contained in the ECMA 7-bit coded character set. Strings are sequences of characters and are used in BASIC programs as comments (see 19), as string constants (see 6), or as data (see 15).

Syntax

letter

A/B/C/D/E/F/G/H/J/K/L/M/N/O/P/Q/R/S/T/U/V/W/X/Y/Z

digit

0/1/2/3/4/5/6/7/8/9

string-character

quotation-mark / quoted-string-character

quoted-string-character

exclamation-mark / number-sign / dollar-sign / percent-sign / ampersand / apostrophe / left-parenthesis / right-parenthesis / asterisk / comma / solidus / colon / semi-colon / less-than-sign / equals-sign / greater-than-sgn / question-mark / circumflex-accent / underline / unquoted-string-character

unquoted-string-character

space / plain-string-character

plain-string-character

plus-sign / minus-sign / full-stop / digit / letter

remark-string

string-character*

quoted-string

quotation-mark quoted-string-character* quotation-mark

unquoted-string

plain-string character / plain-string-character unquoted-string-character* plain-string-character

Examples

“‘ ANY CHARACTERS AT ALL (?!*!!) CAN BE USED IN A “REMARK”.

“SPACES AND COMMAS CAN OCCUR IN QUOTED STRINGS.”

COMMAS CANNOT OCCUR IN UNQUOTED STRINGS.

Semantics

The letters shall be the set of upper-case Roman letters contained in the ECMA 7-bit coded character set in positions 4/1 to 5/10.

The digits shall be the set of arabic digits contained in the ECMA 7-bit code. . .

. . .

THe names of characters are specified in Table 1.

The coding of characters is specified in Table 2; however this coding applies only when programs and/or input/output data are exchanged by means of coded media.

Exceptions

None.

Remarks

. . .

Programs

General Description